STUDENTS' TECHNICAL COUNCIL

BLACK HAWK
REIMAGINED REINVENTED REDEFINED

14-16 APRIL 2023

TECHSHILA

INTER BHAWAN TECH MEET

Drone Robotics

BlackHawk's Drone Swarming Challenge

Drones are a prime example of cutting edge technology that brings together several fields of engineering. Innovations in science and technology bring about revolutionary changes in human life, and drones are no different. The development of drone technology now and in the near future will help India become a superpower in the coming decades. You, the IITians, are given a task today to work on something that has wide-ranging civilian and military applications.

At BlackHawk we strive to achieve various autonomous drone solutions that make drone handling as easy as using a smartphone. For drones to take over in the future, multi-agent drone technology is needed, for which we assign a mission to the hands of India's finest engineers. You are required to solve a problem statement which incorporates swarming among drones to create various patterns.

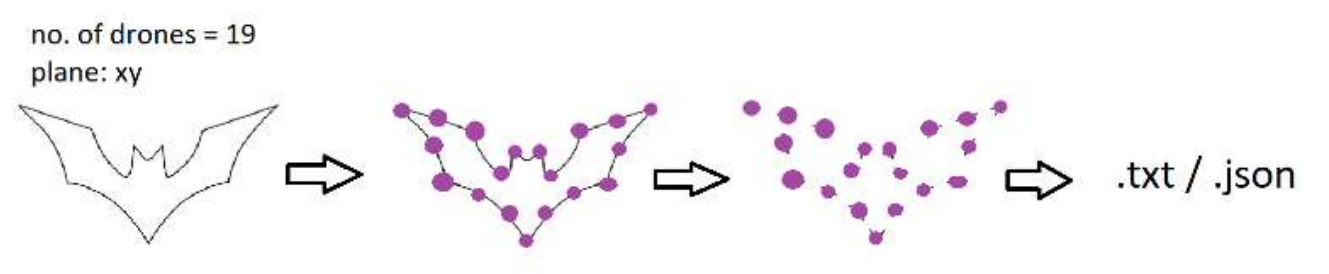
Problem Statement

Task 1: Developing a GUI application to create drone patterns: [45 points]

The objective is to generate a 2D drone pattern from a geometric shape. Create a GUI application to take three inputs:

1. Number of drones (any number from 1-100)
2. The plane on which the drone pattern must be formed (xy, yz or xz planes)
3. Drawing of a geometric shape (either drawn freehand or constructed using lines and arcs)

The application should output an array of coordinates which represent the desired drone positions. These coordinates must be saved to a text/json file.
This can be illustrated as follows:



There is no restriction on the tech stack to be used. Examples of the tech stack include PyQT5, Tkinter and PySimpleGUI libraries for Python, Java, C++ and C#. The only requirement is that it should be a desktop GUI application and NOT a web application.

DELIVERABLES FOR TASK 1:

- Application code (uploaded to a git repository + submission of zip file)
- Clear instructions for installation
- Video of the working application

SCORING FOR TASK 1:

- Accuracy of the generated coordinates [25 points]
- UI/UX [15 points]
- Ease of installation [5 points]

Task 2: Drone Swarming in Robot Operating System (ROS) [40 points]

Simulation files and installation instructions for Task 2 will be provided to teams within 1 day of team registration, so quicker registration will give teams a headstart.

Objective: To read the generated coordinates from the .txt/.json file saved in Task 1 and write python code to control the drones so as to reach those desired coordinates without colliding with other drones. For task 2, the number of drones is set to 12.



2D drone pattern in xy plane with 12 drones

DELIVERABLES FOR TASK 2:

- Scripts for controlling the drones (uploaded to a git repository + submission of zip file)
- Video of the simulation

SCORING FOR TASK 2:

- Accuracy and speed of reaching a known formation (revealed soon after team registration) [20 points: 10 for accuracy and 10 for speed]

- Accuracy and speed of reaching an unknown formation (revealed/tested on the day of evaluation) [20 points: 10 for accuracy and 10 for speed]

Software specifications:

1. ROS for simulation (ROS Noetic on Ubuntu 20.04)
2. Gazebo simulator (Gazebo 11)
3. Ardupilot software in the loop (SITL) drone firmware (version 4.2)

Libraries and frameworks that can be used by teams:

1. Pymavlink
2. MavROS
3. Any other library/framework that can interact with Ardupilot SITL through the Mavlink protocol.

Task 3: Computer Aided Design (CAD) and ANSYS simulation of a quadcopter: [15 points]

DELIVERABLES FOR TASK 3:

- CAD model in Solidworks
- ANSYS simulation
- Brief 1 page report on manufacturing and material usage.

SCORING FOR TASK 3:

- Innovative and practical design with ease of manufacturing [5 points]
- Safety features (e.g. propeller guards that help against lateral collisions) [5 points]
- ANSYS simulation (aerodynamics, COG stability, stress-strain testing) [5 points]

Evaluation Criteria

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- Task 1: 45 Points
- Task 2: 40 Points
- Task 3: 15 Points

Participation

- The Maximum Team Size: 10 members
- Contact your Bhawan's Contingent Leader to participate.
- For any queries reach out to us through our official mail id: BHawkInnovations@gmail.com

We look forward to seeing your submissions and wish you the best of luck!